

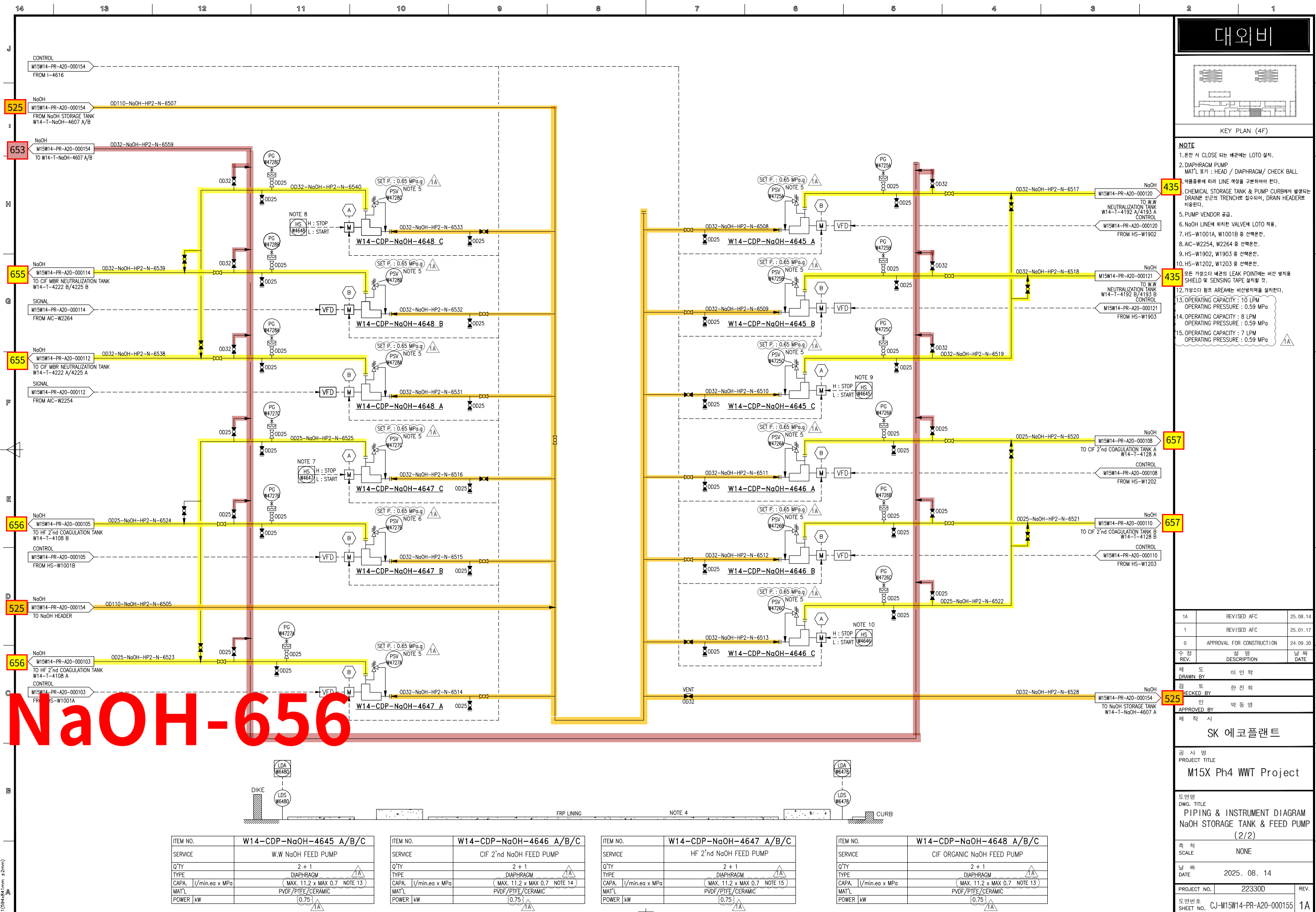
DRAWING LIST FOR TEST PACKAGE

NAOH-656

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	656	4FA	OD25-NaOH-HP2-N-6523-01	1 OF 2	0	CJ-M15W14-PR-A20-000155		
2	656	4FA	OD25-NaOH-HP2-N-6523-02	2 OF 2	0	CJ-M15W14-PR-A20-000155, 103		
3	656	4FA	OD25-NaOH-HP2-N-6524-01	1 OF 2	0	CJ-M15W14-PR-A20-000155		
4	656	4FA	OD25-NaOH-HP2-N-6524-02	2 OF 2	0	CJ-M15W14-PR-A20-000155, 105		
5	656	4FA	OD25-NaOH-HP2-N-6525-01	1 OF 1	0	CJ-M15W14-PR-A20-000155		
6	656	5FA	OD25-NaOH-HP2-N-6523-01	1 OF 1	0	CJ-M15W14-PR-A20-000103		
7	656	5FA	OD25-NaOH-HP2-N-6524-01	1 OF 1	0	CJ-M15W14-PR-A20-000105		
8	656	RFA	OD25-NaOH-HP2-E-6523-01	1 OF 1	0	CJ-M15W14-PR-A20-000103		
9	656	RFA	OD25-NaOH-HP2-E-6524-01	1 OF 1	0	CJ-M15W14-PR-A20-000105		
10	656	4FA	OD32-NaOH-HP2-N-6559-01	1 OF 2	0	CJ-M15W14-PR-A20-000155		
11								
12								



NaOH-656

대외비

KEY PLAN (4F)

NOTE

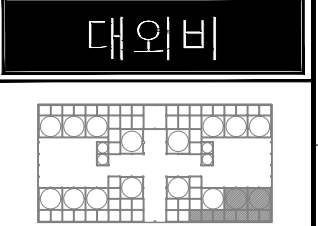
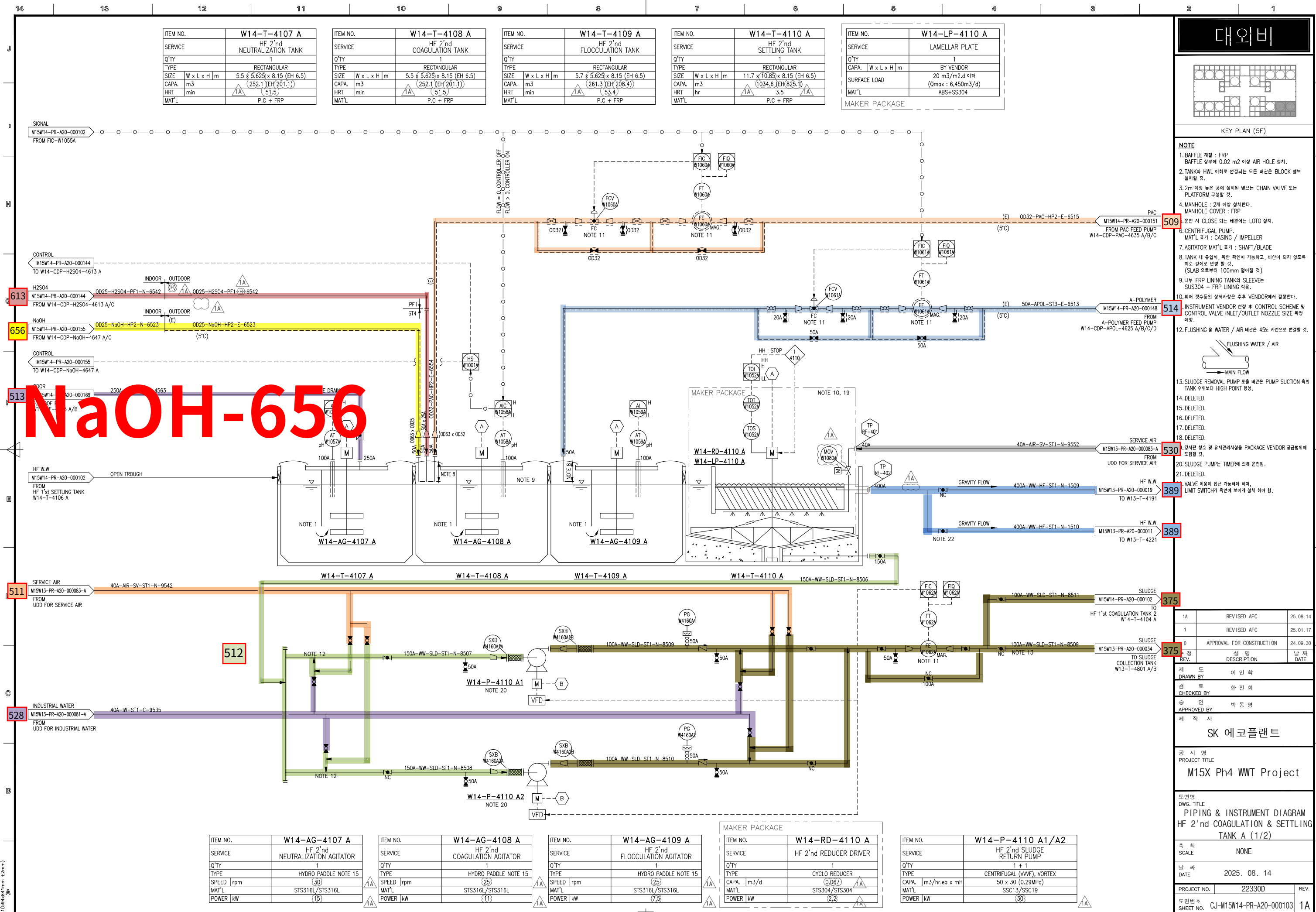
1. 운전 시 CLOSE 되는 배관에는 LOTO 실시.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 역류종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 규격.
6. NaOH LINE에 위치한 VALVE에 LOTO 적용.
7. HS-W1001A, W1001B 중 선택은전.
8. AIC-W2254, W2264 중 선택은전.
9. HS-W1902, W1903 중 선택은전.
10. HS-W1202, W1203 중 선택은전.

모든 가성소터 배관의 LEAK POINT에는 비산 방지를 SHIELD 및 SENSING TAPE 설치할 것.

12. 가성소터 배관 AREA에는 비산방지막을 설치한다.
13. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.59 MPa
14. OPERATING CAPACITY : 8 LPM
OPERATING PRESSURE : 0.59 MPa
15. OPERATING CAPACITY : 7 LPM
OPERATING PRESSURE : 0.59 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인 학
DRAWN BY		
검	토	한 진 희
CHECKED BY		
APPROVED BY	박 동 영	
제	작	사
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
NaOH STORAGE TANK & FEED PUMP		
(2/2)		
속		
척		
SCALE		
NONE		
날		
짜		
DATE		
2025. 08. 14		
PROJECT NO.		
22330D		
REV.		
A		
도면번호		
SHEET NO.		
CJ-M15W14-PR-A20-000155		
1A		

ITEM NO.	W14-CDP-NaOH-4645 A/B/C	ITEM NO.	W14-CDP-NaOH-4646 A/B/C	ITEM NO.	W14-CDP-NaOH-4647 A/B/C	ITEM NO.	W14-CDP-NaOH-4648 A/B/C
SERVICE	W.W NaOH FEED PUMP	SERVICE	CIF 2'nd NaOH FEED PUMP	SERVICE	HF 2'nd NaOH FEED PUMP	SERVICE	CIF ORGANIC NaOH FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa	CAPA.	1/min.ea x MPa
MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC	MAT'L	PVDF/PTFE/CERAMIC
POWER	kw	POWER	kw	POWER	kw	POWER	kw



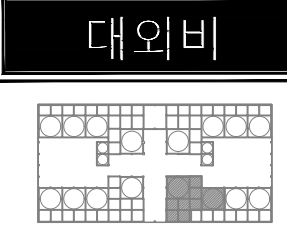
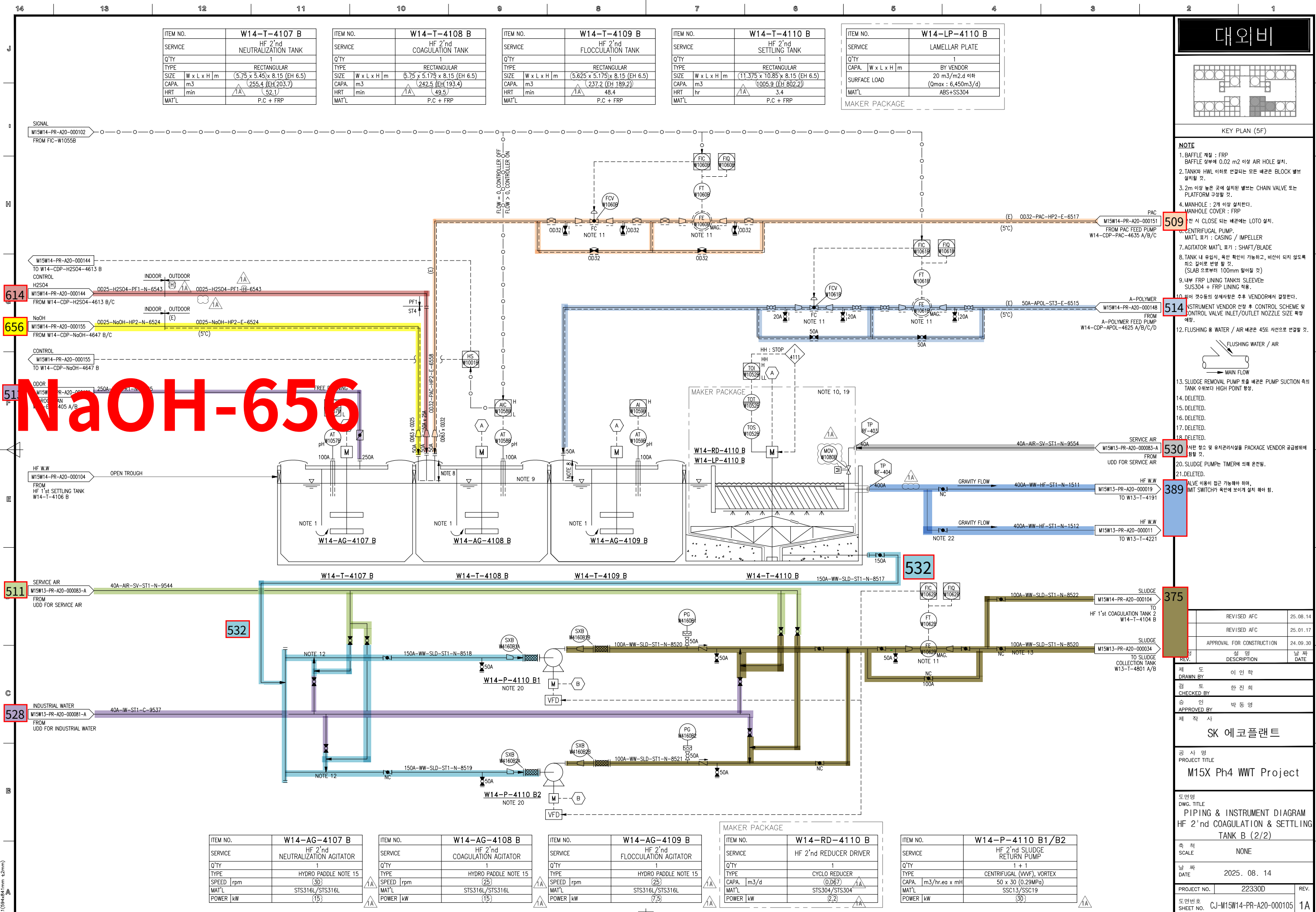
- NOTE
1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 거주등의 상해사상은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향함.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 경사한 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. VALVE 이용이 접근 가능해야 하며, LIMIT SWITCH가 육안에서 보이게 설치 해야 함.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
REV.	실명	날짜
DESCRIPTION	DATE	

DRAWN BY 이인학
CHECKED BY 한진희
APPROVED BY 박동명
제작사

SK 에코플랜트
공사업
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
HF 2'nd COAGULATION & SETTLING TANK A (1/2)
속척
SCALE
NONE
날짜
DATE
2025. 08. 14
PROJECT NO.
22330D
REV.
도면번호
SHEET NO.
CJ-M15W14-PR-A20-000103
1A



KEY PLAN (5F)

- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 전 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB 으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 제어 시스템의 상세사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서면으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 할 것.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 서면 참조 및 유지관리시도를 PACKAGE VENDOR 공급범위에 포함 할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.
 22. ALVE 이용이 접근 가능해야 하며, MIT SWITCH가 육안에 보이게 설치 해야 함.

5		REVISED AFC	25.08
		REVISED AFC	25.01
		APPROVAL FOR CONSTRUCTION	24.05
	당	실 명	날

공사업	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM	
HF 2'nd COAGULATION & SETTLING TANK B (2/2)	
속척	SCALE
NONE	
날짜	DATE
2025. 08. 14	

PROJECT NO.	22330D	REV.	A
도면번호	CJ-M15W14-PR-A20-000105		
SHEET NO.	1A		

ITEM NO.	W14-T-4107 B
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.45 x 8.15 (EH 6.5))
CAPA.	m3 (255.4 (EH 203.7))
HRT	min (52.1)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 B
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.75 x 5.175 x 8.15 (EH 6.5))
CAPA.	m3 (242.5 (EH 193.4))
HRT	min (49.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 B
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (5.625 x 5.175 x 8.15 (EH 6.5))
CAPA.	m3 (237.2 (EH 189.2))
HRT	min (48.4)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 B
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m (11.375 x 10.85 x 8.15 (EH 6.5))
CAPA.	m3 (1005.9 (EH 802.2))
HRT	hr 3.4
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 B
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304

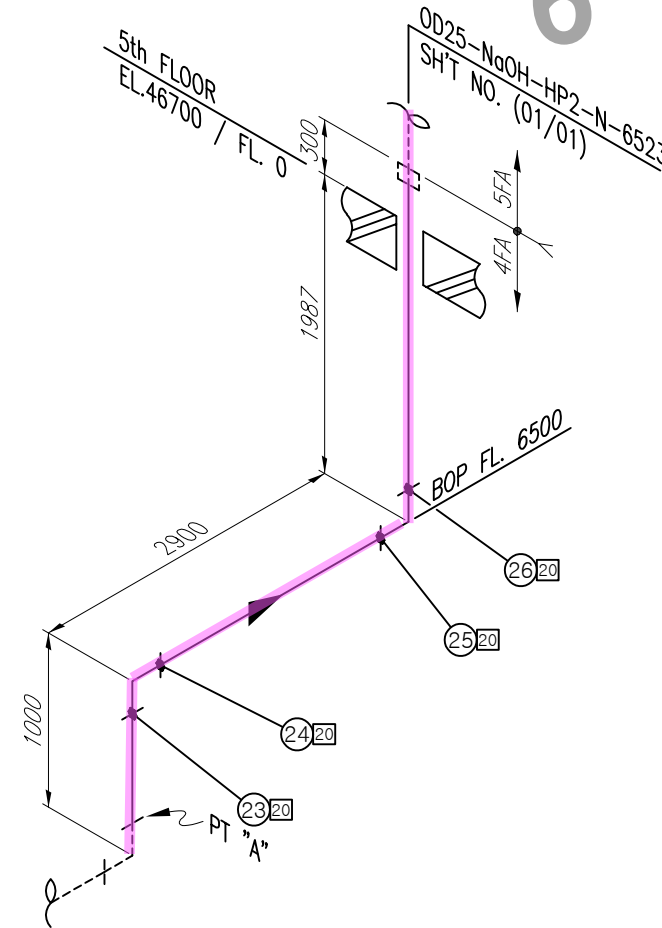
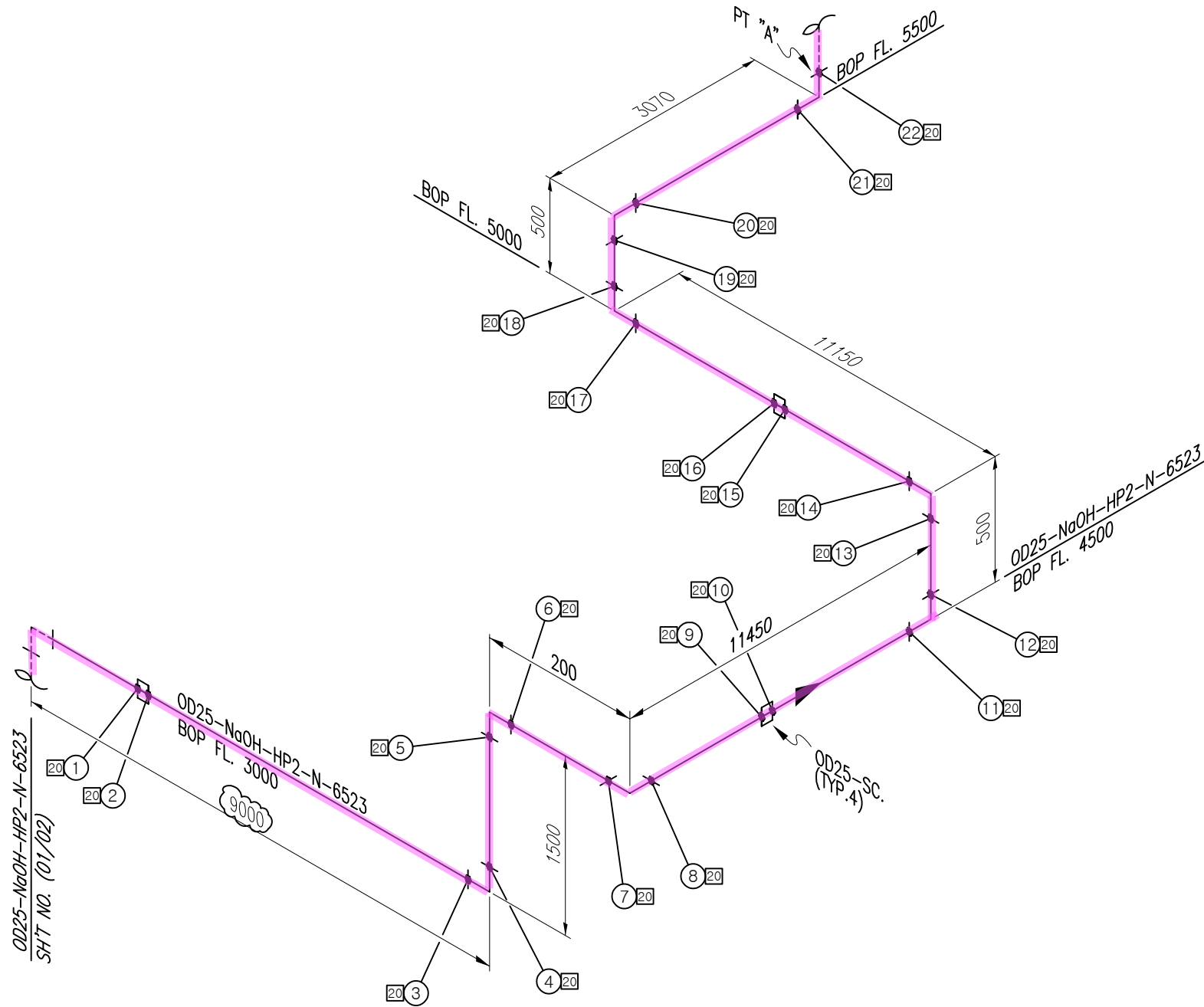
ITEM NO.	W14-AG-4107 B
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4108 B
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (15)

ITEM NO.	W14-AG-4109 B
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-RD-4110 B
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4110 B1/B2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)

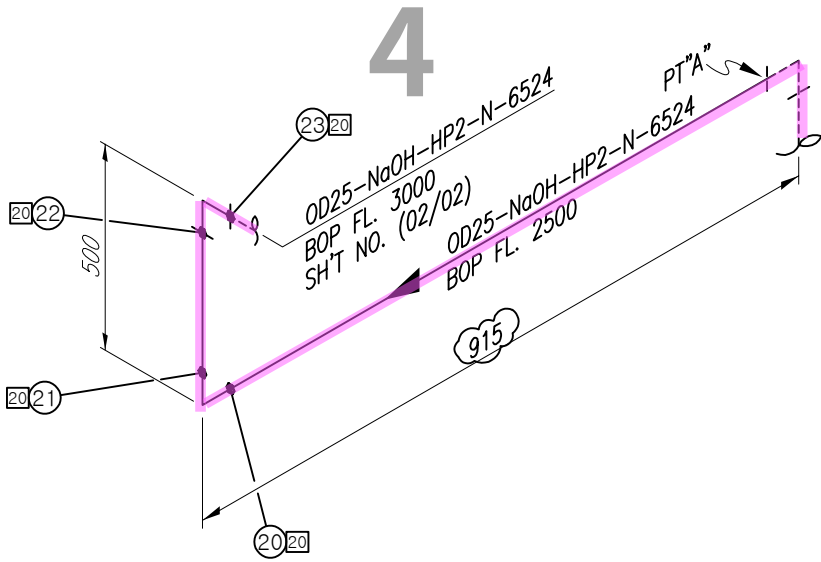
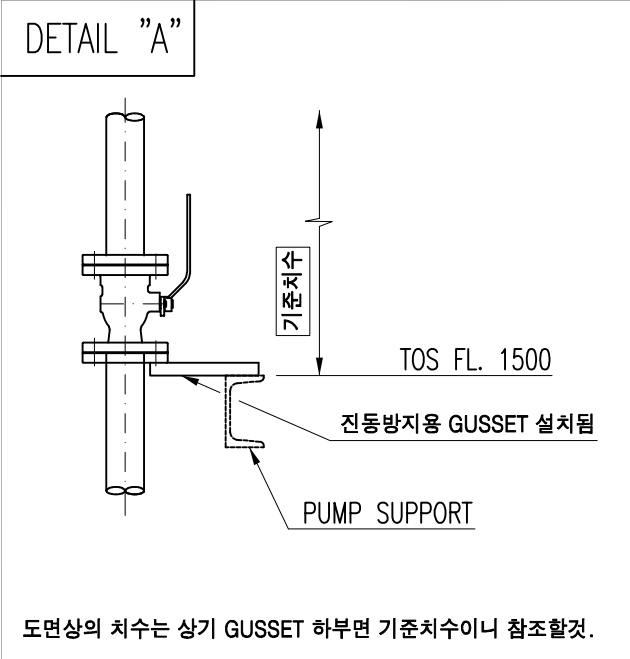
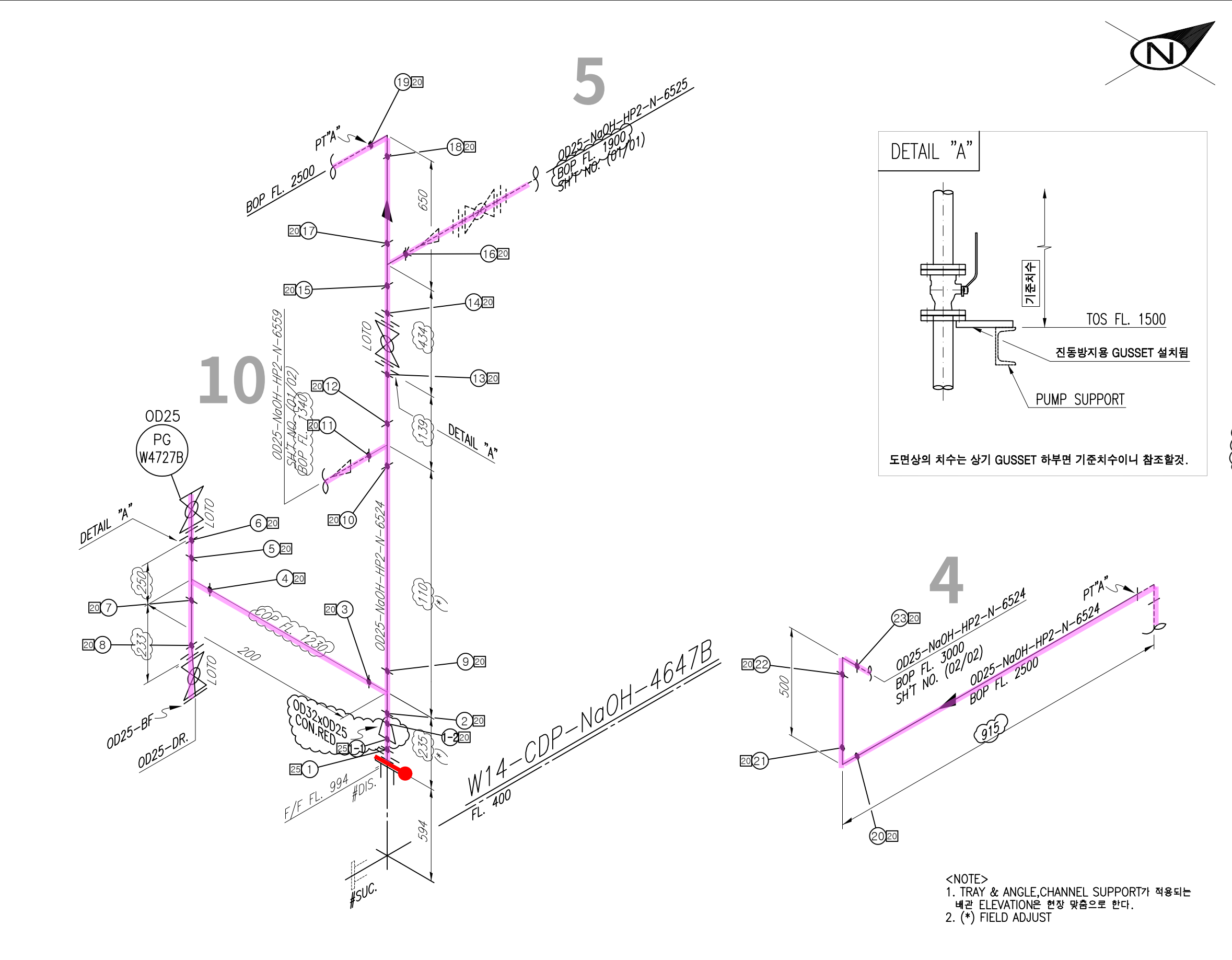


1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

2

SH'T NO. 02 / 02	REV. 1
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정고: 본 문면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	3.3M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD25	3EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	4EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	3EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
SH	OD25	6EA	SEMI-PP / SEAT-PTFE, FB, LEVER + LOTO
PG	OD25	1EA	PRESSURE GAUGE, KS 10K, FF

RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD32	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
G	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B1	OD32	1SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

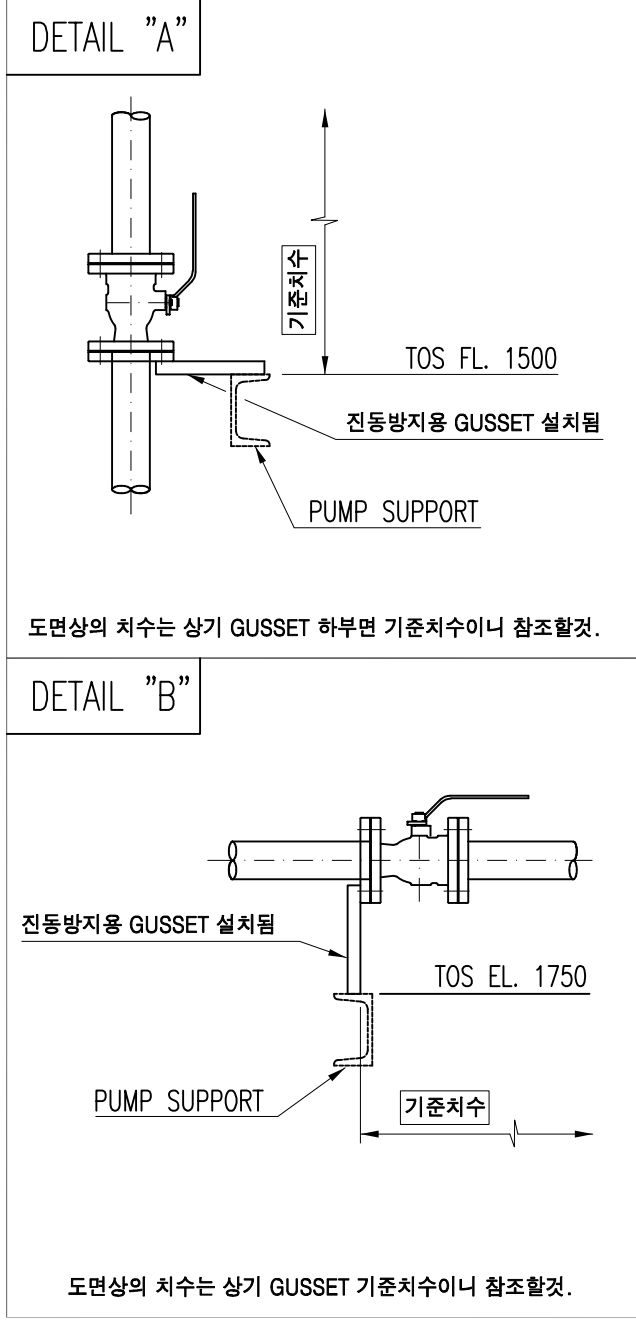
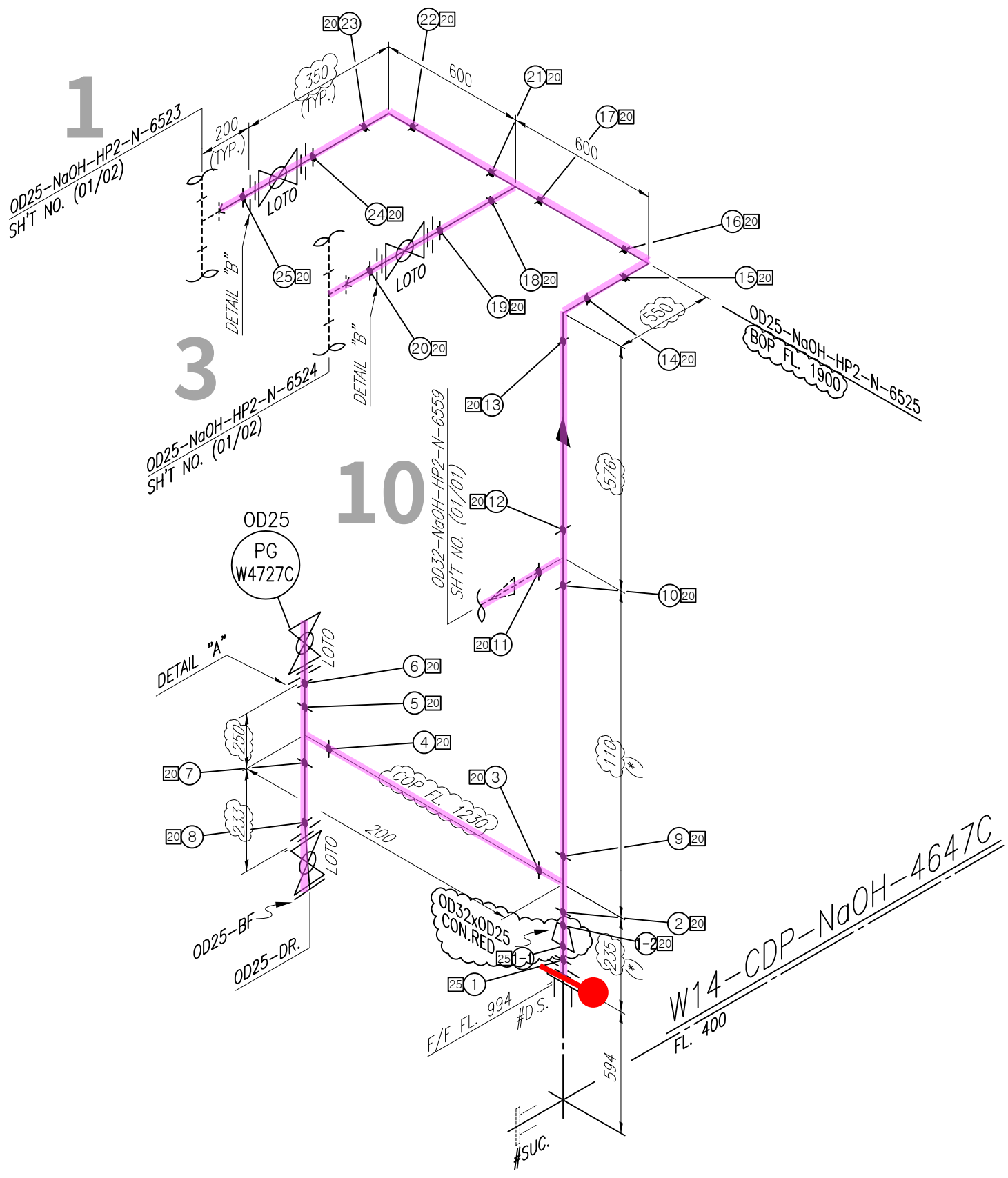
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	6.8	60	UNIT	4FA										
			OPERATING	6.1	AMB	REF. PLAN DWG.										SK ecoplant	M15X Ph4 Project
						CJ-M15W14-PI-A30-040009-A										COMMODITY	NaOH
						REF. P & ID										LINE NO.	OD25-NaOH-HP2-N-6524
						CJ-M15W14-PR-A20-000155										SH'T NO.	01 / 02
																REV.	1



4

CLIENT  4

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있습니다.



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	4.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD25	3EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION,
T	OD25	4EA	TEE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	6EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-ST304 / STUB END-HDPE PE100
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	6EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	7SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	4EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	8EA	SHIELD
PG	OD25	1EA	PRESSURE GAUGE, KS 10K, FF

RC	OD32xOD25	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD32	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
G	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B1	OD32	1SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR		PROJECT NAME		
	THICK.	—	DESIGN	6.8	60	UNIT	4FA									M15X Ph4 Project		
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ—M15W14—PI—A30—040012—A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY		JOB NO.		
						REF. P & ID CJ—M15W14—PR—A20—000155								NaOH		22330D		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.		OD32—NaOH—HP2—N—6525	SH'T NO. 01 / 01	REV. 0

[illegible]

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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]

※ 참고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

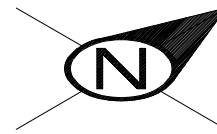
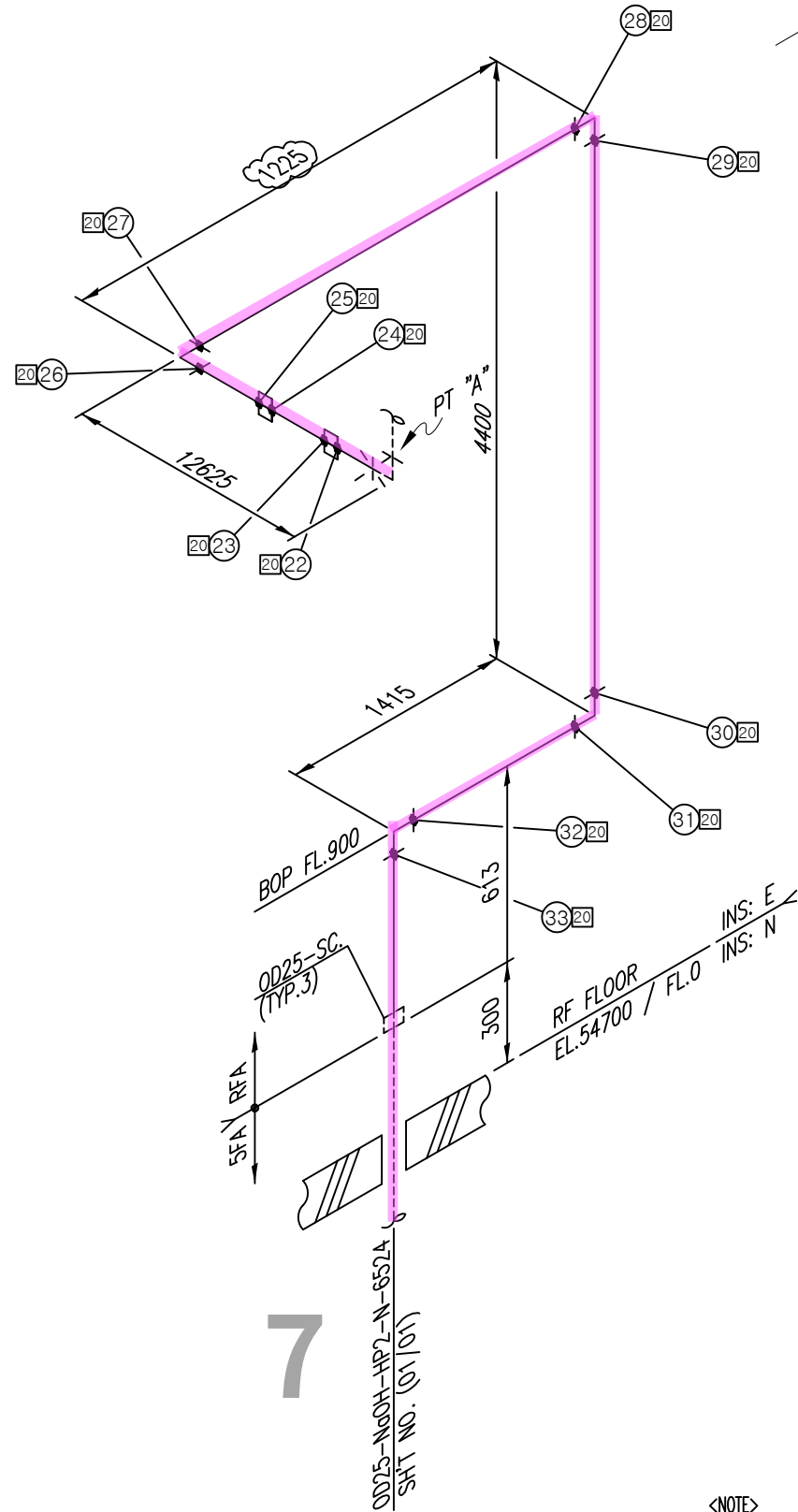


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	CONTRACTOR		PROJECT NAME	
	COMMODITY	NaOH	JOB NO.	
영			22330D	
D	LINE NO.	OD25-NaOH-HP2-N-6524	SH'T NO.	REV.
			01 / 01	0

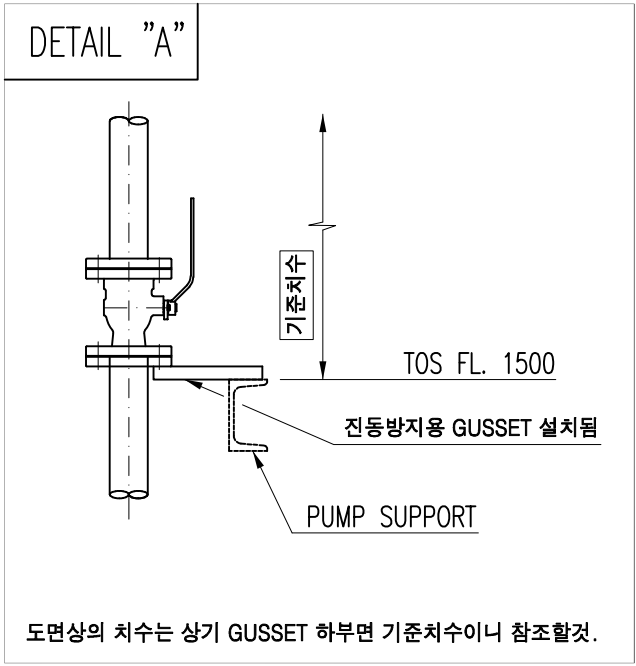
[illegible]

SK HYNIX의 기업정보를 제공한 자료로서 어떠한 이유에서든 본 제3자에게 공개하거나 누락·변경·왜곡·소홀한 사실이나 허위사실을 포함하여 다른 당사자에게 인민경제생활에 영향을 미칠 수 있는



CLIENT  9

※고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 사전의 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생한 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

[illegible]

CLIENT



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CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY NaOH		JOB NO.	
		22330D	
LINE NO.	OD32-NaOH-HP2-N-6559	SH'T NO.	REV.
		01 / 02	0